

ROBERTO GUAGNELLI GARCÍA

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Mechanical Engineering Student | Overall Aircraft Design & Systems Integration

Spanish & Mexican citizen | EU work authorization | Open to relocation

EDUCATION

Tecnológico de Monterrey, Campus Estado de México

Expected Jul 2027

B.S. in Mechanical Engineering; Concentration in Aeronautics

Academic average: 92/100

WORK EXPERIENCE

GE Aerospace | Designer Intern, Product Definition

Aug 2026 – Present

RISE Open Fan Program

- Design mounting brackets and update electrical harness routing in Siemens NX; completed 70+ brackets and 20+ routing revisions while applying manufacturing and maintainability criteria to mechanical/electrical integration.
- Identify and resolve clearance and minimum bend radius noncompliances by designing new harness support brackets; review component interferences and routing requirements.
- Prepare product definition drawings using tolerance tables and established datums from internal design practices for brackets and electrical harnesses on the RISE Open Fan program.

GE Aerospace | Lean Summer Experience Intern

Jun 2026 – Jul 2026

Integrated Performance & Operability (IP&O)

- Implemented visual planning, KPI dashboards and weekly review practices with subsection managers and program leads; reduced average meeting duration by 20%, as timed by meeting leaders.

ENGINEERING PROJECTS & LEADERSHIP

Borregos CEM AeroDesign | Tecnológico de Monterrey

2024 – Present

Engineering Coordinator / Design Captain

Aug 2024 – Sep 2026

- Lead a five-person design team and coordinate overall aircraft design and systems integration across structures, electronics, design and R&D within a 40+ member team; document requirements and conduct multidisciplinary design reviews.
- Resolved a structural/electrical interface conflict between a continuous structural part and cable passage needs; coordinated geometry and cabling changes across both teams.
- Modeled 300+ CAD parts per aircraft within 400+ component assemblies; performed aircraft configuration and physical-integration trade studies covering mass, CG, stability, propulsion and geometric alternatives.
- Designed removable joints for Wingull's 4.2 m wing to fit components under 1.2 m; supported an 18 kg distributed-mass verification test with tip deflection within 8% of the structural team's prediction.
- Performed steady-state/transient CFD and XFLR5/AVL analyses covering aerodynamic performance, static stability, flight mechanics and flight dynamics; assessed trim and elevator/aileron/rudder control authority using stability and control derivatives, and used drag coefficients in takeoff models to estimate allowable payload.

Manufacturing Coordinator | Hyperion

2024

- Coordinated eight members in fabrication, assembly and quality control; designed and manufactured a welded-aluminum nose landing gear with a spring, travel stop and fork to accommodate limited propeller clearance.
- Helped build an aircraft that completed six competition flights carrying 3 kg payload at 4 kg empty mass; team placed 2nd overall and 3rd in design report at SAE Aero Design México 2024.

TECHNICAL SKILLS

CAD & Analysis: Siemens NX, SolidWorks, Fusion 360, ANSYS Fluent, XFLR5, AVL; steady-state and transient CFD.

Programming & Data: Python, MATLAB, Excel, Power BI.

Aircraft & Systems: Overall aircraft design, systems integration, physical integration, flight physics, flight mechanics, flight dynamics, aircraft configuration, propulsion selection, requirements documentation, interface management, verification/testing.

Stability & Control: Static stability, trim analysis, control-surface effectiveness, control derivatives, control-authority assessment using AVL/XFLR5.

Manufacturing: 3D printing, carbon-fiber fabrication, laser cutting, aluminum welding, tube and sheet bending.

Certification: Certified SOLIDWORKS Professional (CSWP), perfect exam score.

Languages: Spanish (native), English (C2; Cambridge and IELTS), German (currently learning at A2 level).